

Question #1 of 120

Question ID: 1214789

Questions 1 through 18 relate to Ethical and Professional Standards. (27 minutes)

Adam Schute, CFA, is on a conference call with the CFO of an investment banking client with his phone speaker on and his door open. As a result, salesmen and traders overhear the CFO describing problems with production target dates that have not been publicly disclosed. The salesmen relay this information to clients and the traders reduce their positions in the stock. With respect to the Standard on material nonpublic information, Schute has:

- A) not violated the Standard because he has not acted on the information, but the traders and salesmen have violated the Standard.
- B) violated the Standard because he should have taken steps to prevent the dissemination of the information.**
- C) violated the Standard by not making the information public when he realized others had overheard the call.

Explanation

Schute has violated the Standard by not taking steps to ensure that the nonpublic information he has received as part of his investment banking work was not shared with others in the firm.

For Further Reference:

(Study Session 1, Module 3.3, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #2 of 120

Question ID: 1214800

Peter Wellington has changed his status in marketing materials to "Level III CFA candidate." Wellington passed the Level II CFA exam and just received his results. He intends to register for the next Level III CFA examination that is offered the following June. Wellington has:

- A) not violated the Standards of Professional Conduct.
- B) violated the Standard on conduct as participants in CFA Institute programs.
- C) violated the Standard on reference to the CFA Institute, the CFA Designation, and the CFA Program.**

Explanation

Under Standard VII(B) Reference to CFA Institute, the CFA Designation, and the CFA Program, candidates in the CFA Program may appropriately reference their participation in the CFA Program. However, to be considered a candidate, individuals must be registered to

take the next scheduled CFA examination. Therefore, Wellington, who intends to register for the next exam, but has not done so, has violated Standard VII(B). Wellington may indicate that he passed Level II of the CFA Program and the year in which he passed, but he is not a candidate unless he is registered for the next exam or awaiting exam results. Standard VII(A) Conduct as Participants in CFA Institute Programs addresses issues that concern the validity and security of the exams and the integrity and reputation of the CFA designation and CFA Institute.

For Further Reference:

(Study Session 1, Module 3.9, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #3 of 120

Question ID: 1214802

Which of the following statements relating to the Global Investment Performance Standards (GIPS®) is *least accurate*?

- A) Only investment management firms may claim compliance with GIPS.
- B) GIPS represent standards to which members of CFA Institute and CFA candidates must adhere.**
- C) To claim GIPS compliance, a firm must present at least five years (or since its inception if less than five years) of annual investment performance that complies with GIPS.

Explanation

Global Investment Performance Standards represent ethical reporting standards, but compliance with GIPS is not a requirement of CFA Institute membership or to sit for the CFA examination.

For Further Reference:

(Study Session 1, Module 4.1, LOS 4.a)

CFA® Program Curriculum, Volume 1, page 227

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Question ID: 1214794

When he assumed the job of compliance officer two years ago, Ed Michaels, CFA, issued written compliance procedures and made all covered employees aware of the procedures. A report by an external auditor found that on several occasions over the past two years, two different employees traded in recommended securities ahead of trades made in managed client accounts. Michaels fires both employees and recirculates the written compliance procedures that explain clearly which activities are prohibited. Michaels has violated the Standard concerning:

- A) Responsibilities of Supervisors by firing the employees instead of restricting their activities.
- B) Responsibilities of Supervisors by failing to implement reasonable procedures to detect violations.**
- C) Misconduct because, as the compliance officer, he is associated with, and ultimately responsible for, the unethical activity.

Explanation

Michaels has violated Standard IV(C) Responsibilities of Supervisors, which requires him to make reasonable efforts to detect and prevent violations of compliance procedures. Simply making employees aware of the rules is not enough. Monitoring of employee

trades, duplicate confirms, establishing blackout periods, or preclearance of employee trades are all methods that would have revealed the problem prior to the external audit. Michaels has not violated Standard I(D) Misconduct because his actions did not exhibit dishonesty or lack of integrity.

For Further Reference:

(Study Session 1, Module 3.2, 3.6, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #5 of 120

Question ID: 1214797

Ralph Malone, CFA, has many clients, including a trust account that benefits three of his immediate family members. His firm changes its recommendation on a stock from "hold" to "buy" on a security that is suitable for many clients, including the trust. Which of the following would be considered a violation of the Standard concerning priority of transactions?

- A) Malone waits until after the firm purchases the security to buy it for the family trust account.**
- B) Malone trades on the family account shortly after his firm's clients have been informed of the buy recommendation.**
- C) The firm gives clients time to act on the new recommendation and then buys 100,000 shares for its own account prior to publicly disclosing the new recommendation.**

Explanation

Because the family account is a client account, it should be treated as any other client account would be. Waiting until after the firm buys shares would violate Standard VI(B) Priority of Transactions. There is no requirement that a firm's recommendations be made public.

For Further Reference:

(Study Session 1, Module 3.8, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

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Question ID: 1214795

Patricia Nelson, CFA, is informed by one of her clients that if she can get the performance of the client firm's pension portfolio above that of the Standard & Poor's average by year-end, the client will give her a free trip to Singapore to visit the firm's offices. If Nelson agrees to this arrangement, which of the following actions complies with CFA Institute Standards of Professional Conduct? Nelson:

- A) must inform her employer of this agreement but does not need consent.**
- B) must inform her employer of this agreement and may accept it with verbal consent.**
- C) may inform her employer by email of this agreement and must receive written consent.**

Explanation

To comply with Standard IV(B) Additional Compensation Arrangements, because the additional compensation is contingent on future performance, Nelson must disclose this additional compensation to her employer and must receive written consent, which can be email or any other form of communication that can be documented.

For Further Reference:

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Question ID: 1214796

John Anderson, CFA, follows Radley Manufacturing for his employer, Atlas Brokers. Radley has announced a takeover bid for Palmer Industries, a company that Atlas does not follow. To get out a research note about the potential effects of the acquisition on Radley, Anderson purchases a report on Palmer from a research firm Atlas regularly uses. Anderson incorporates projections and analysis from the purchased report into his research note on what a post-acquisition Radley would look like but does not cite the source. Anderson has violated the Standard on:

- A) misrepresentation.**
- B)** independence and objectivity.
- C)** diligence and reasonable basis.

Explanation

Anderson has violated Standard I(C) Misrepresentation by representing the work of another as his own in the report. The use of third-party research is not necessarily a violation of Standard I(B) Independence and Objectivity or Standard V(A) Diligence and Reasonable Basis. Using a research firm that is approved by his company is permitted as long as he has no knowledge that the work is of less than good quality.

For Further Reference:

(Study Session 1, Module 3.1, 3.2, 3.7, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

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Question ID: 1214786

Roger Smith, CFA, has been invited to join a group of analysts in touring the riverboats of River Casino Corp. For the tour, River Casino has arranged chartered flights from casino to casino since commercial flight schedules are not practical for the group's time schedule. River Casino has also arranged to pay for the analysts' lodging for the three nights of the tour. According to CFA Institute Standards of Professional Conduct, Smith:

- A) may accept the arrangements as they are.**
- B)** may accept the flight but is required to pay for his lodging.
- C)** is required to pay for his flight and lodging.

Explanation

Because the itinerary required charter flights due to a lack of commercial transportation, River Casino can appropriately provide them. While Standard I(B) Independence and Objectivity recommends that members pay their own room costs, it is not required and it is not unusual for members to accept accommodations.

For Further Reference:

(Study Session 1, Module 3.1, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #9 of 120

Question ID: 1214803

When calculating the performance of a composite, which of the following actions complies with GIPS?

- A) Defining a composite so as to include portfolios that have been discontinued.**
- B)** Assigning each portfolio to a single composite at the end of the year based on its holdings over the year.
- C)** Calculating a composite's return as the mean annual total return of portfolios assigned to the composite.

Explanation

GIPS require discontinued portfolios to be included in composites. To comply with GIPS, portfolios must be assigned to composites before the returns are known; assigning them at the end of the year is not acceptable. The composite return must be asset weighted, not a simple average (equal-weighted).

For Further Reference:

(Study Session 1, Module 4.1, LOS 4.b, Module 5.1, LOS 5.a)

CFA® Program Curriculum, Volume 1, page 229

CFA® Program Curriculum, Volume 1, page 235

Question #10 of 120

Question ID: 1214798

Susan Smart, CFA, is about to change her "buy" recommendation on RollinsCo to "sell." RollinsCo had been growing rapidly over the past year, but Smart believes the growth potential is now gone. Smart sells the shares held in her discretionary client accounts and in her own personal account before issuing her report. According to the Standards that concern fair dealing and priority of transactions, Smart violated:

- A) both of these Standards.**
- B)** neither of these Standards.
- C)** only one of these Standards.

Explanation

Smart violated both Standards. Smart violated Standard III(B) Fair Dealing because she did not deal fairly and objectively with all clients and prospects when disseminating investment recommendations, giving priority to some of the firm's clients by trading for her clients first before issuing the report. She also sold her own shares before issuing the report, which violated Standard VI(B) Priority of Transactions. Smart did not give clients an opportunity to react to and benefit from her recommendation before she personally benefited from her research.

For Further Reference:

(Study Session 1, Module 3.4, 3.8, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #11 of 120

Question ID: 1214790

Matt Jacobs, CFA, recommended to a client that he buy shares in Timeco, which has subsequently underperformed the market. Timeco stock is thinly traded, and its price has decreased sharply over the past few weeks because two insiders have sold large blocks of shares. Jacobs believes this price decrease reflects an illiquid market. Because he still believes Timeco is a good long-term investment, he buys shares for his personal account in order to raise the price and help him convince his client to hold on to his investment in Timeco. Has Jacobs violated the Standards?

- A) No, because the trade is for the benefit of his client.
- B) Yes, because he intended to manipulate the market price of Timeco.**
- C) No, because he is providing liquidity to offset the effects of the recent large insider sales.

Explanation

Whatever his motivation, Jacobs' attempt to manipulate the market price of Timeco shares with the intent to deceive market participants (in this case, his own client) constitutes a violation of Standard II(B) Market Manipulation.

For Further Reference:

(Study Session 1, Module 3.3, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #12 of 120

Question ID: 1214791

William Rex, CFA, has a one-man firm that manages investment portfolios. In his marketing materials, Rex presents the asset-weighted returns for accounts he has managed over the last five years and does not disclose that the first two years of his performance history were achieved at a previous firm. He also includes simulated results of a stock selection model he employs and indicates that they are from a simulation. Has Rex violated any CFA Institute Standards of Professional Conduct?

- A) No.
- B) Yes, failing to disclose that two years of his performance results were with a previous employer is a violation, but including the simulated results is acceptable.**
- C) Yes, both by failing to disclose that two years of his performance results were with a previous employer and by including the simulated, rather than actual, performance results.

Explanation

Under Standard III(D) Performance Presentation, members and candidates must make reasonable efforts to ensure that investment performance information is fair, accurate, and complete. Rex has misled his potential clients by not disclosing that the first two years of his performance record were achieved at another firm. There is no prohibition against presenting simulated results as long as the fact that the results are simulated, rather than actual, is disclosed.

For Further Reference:

(Study Session 1, Module 3.5, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #13 of 120

Question ID: 1214787

Tony Roberts, CFA, is part of a team that manages equities accounts. He believes that a teammate, who is not a CFA Institute member or candidate, takes actions that, while not illegal under local law, violate CFA Institute Standards of Professional Conduct. According to

the CFA Institute Standards of Professional Conduct, Roberts:

- A) is required to dissociate from the team's activities if they continue.**
- B)** is not required to act because the Code and Standards do not apply to non-members.
- C)** must report the suspected violations of the Code and Standards first to his supervisor and then to CFA Institute.

Explanation

If Roberts suspects someone is engaging in activities that are illegal or violate the Code and Standards, Standard I(A) Knowledge of the Law requires him to dissociate from the activities if he cannot remedy the situation. In this situation, the teammate is acting within the applicable laws but is violating CFA Institute Standards of Professional Conduct. When the Code and Standards are stricter than applicable law, the Code and Standards apply to members and candidates. However, Roberts is not required by the Code and Standards to report violations of laws or the Code and Standards to CFA Institute or to governmental regulators, although it may be prudent or even required by law that he do so.

For Further Reference:

(Study Session 1, Module 3.1, LOS 3.a, 3.b, 3.c)

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Question #14 of 120

Question ID: 1214788

Judy Blush is a CFA candidate and is recommending the purchase of a mutual fund that invests solely in long-term U.S. Treasury bonds (T-bonds) to one of her clients. She states: "Because the U.S. government guarantees payment of both principal and interest on its bonds, risk of loss from investing in this fund is virtually zero." Blush's actions violated:

- A) the Standard on misrepresentation.**
- B)** the Standard on communication with clients.
- C)** none of the Standards of Professional Conduct.

Explanation

Government bonds are default risk free but are subject to price risk, particularly if the bonds are longer-term than the investor's horizon. Thus, Blush misrepresented the nature of investing in the fund and therefore violated Standard I(C) Misrepresentation.

For Further Reference:

(Study Session 1, Module 3.2, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #15 of 120

Question ID: 1214801

Which of the following actions is *least likely* a violation of the Standard concerning conduct as participants in CFA Institute programs?

- A) A member anonymously posts a disparaging comment about CFA Institute policies on an internet message board.**
- B)** A member fails to disclose a formal complaint from a client on her annual Professional Conduct Statement.

- C)** A candidate discusses which topics were emphasized on the June Level I CFA exam with a candidate for the December exam.

Explanation

Standard VII(A) Conduct as Participants in CFA Institute Programs does not prohibit members and candidates from expressing negative opinions about CFA Institute or the CFA program. Disclosing exam content and misrepresenting information on the annual PCS are violations of Standard VII(A).

For Further Reference:

(Study Session 1, Module 3.9, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #16 of 120

Question ID: 1214792

According to the Standard concerning suitability, it is *most likely* that members and candidates in advisory relationships with clients should:

- A) document unsuccessful attempts to update client information.**
- B)** purchase only securities with low to medium risk for a client with moderate risk tolerance.
- C)** decline to manage assets for clients who withhold important information about their financial circumstances and needs.

Explanation

Members and candidates who are in advisory relationships with clients should document unsuccessful attempts to update client information and circumstances. Because risk is to be evaluated in a portfolio context, there is no prohibition against the purchase of a security that is risky on a stand-alone basis, as long as the risk of the client's overall portfolio is consistent with their ability and willingness to assume investment risk. While lack of client information could make suitability a difficult question, Standard III(C) Suitability does not prohibit managing assets for clients who withhold information about their financial circumstances and needs.

For Further Reference:

(Study Session 1, Module 3.5, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #17 of 120

Question ID: 1214799

William Rogers, CFA, is a commercial insurance broker who sometimes recommends money managers to his high net worth clients. For those clients who hire the managers, the managers pay Rogers a percentage of the management fees on the account. Rogers tells prospects, "I receive referral fees from the money managers if you employ them." Rogers has:

- A)** not violated the Standards.
- B) violated the Standard concerning referrals.**
- C)** violated the Standard on communications with clients and prospects.

Explanation

Standard VI(C) Referral Fees requires members and candidates to disclose any referral fees they will earn from successful referrals as well as the nature of the compensation. Here Rogers must disclose that he will receive a percentage of management fees on an ongoing basis.

For Further Reference:

(Study Session 1, Module 3.8, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #18 of 120

Question ID: 1214793

Byron Bell, CFA, tells his assistant that Mary Mitchel, a client of his, confided to him that she is suffering from the early stages of Alzheimer's disease and that she is planning to leave almost all of her sizable estate to Prather House, a support facility for Alzheimer's patients. Bell directs his assistant to keep this information confidential. With respect to the Standard on preservation of confidentiality, Bell has:

- A) not violated the Standard.**
- B) violated the Standard by sharing information about his client's health but not about planned bequest to Prather House.**
- C) violated the Standard both by sharing the information about his client's health and about her planned bequest to Prather House.**

Explanation

Bell is permitted to share confidential client information with another firm employee who is also working for the client's benefit, so sharing this information with his assistant is not a violation of Standard III(E) Preservation of Confidentiality.

For Further Reference:

(Study Session 1, Module 3.5, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #19 of 120

Question ID: 1214804

Questions 19 through 30 relate to Quantitative Methods. (18 minutes)

A client plans to retire in 15 years and will need to withdraw \$50,000 from his retirement account each year for 10 years, beginning on the day he retires. After that, he will need to withdraw \$20,000 per year for 25 years. The account returns 4% annually. The amount he needs to have in the account on the day he retires is *closest* to:

- A) \$580,000.**
- B) \$640,000.**
- C) \$655,000.**

Explanation

We can treat these cash flows as a 35-year annuity due of \$20,000 per year plus a 10-year annuity due of an additional \$30,000 per year. With calculator in BGN mode:

\$20,000 per year for 35 years: $N = 35$, $PMT = 20,000$, $I/Y = 4$, $FV = 0$; CPT $PV = -388,224$

\$30,000 per year for 10 years: $N = 10$, $PMT = 30,000$, $I/Y = 4$, $FV = 0$; $CPT PV = -253,060$

Total: \$641,284

For Further Reference:

(Study Session 2, Module 6.2, LOS 6.f)

CFA® Program Curriculum, Volume 1, page 316

Question #20 of 120

Question ID: 1214805

A company reports its past six years' earnings growth at 10%, 14%, 12%, 10%, –10%, and 12%. The company's average compound annual growth rate of earnings is *closest* to:

- A) 7.7%.
- B) 8.0%.
- C) 8.5%.

Explanation

Geometric mean = $[(1.10)(1.14)(1.12)(1.10)(0.90)(1.12)]^{1/6} - 1 = 0.0766$, or 7.66%

For Further Reference:

(Study Session 2, Module 7.2, LOS 7.e)

CFA® Program Curriculum, Volume 1, page 386

Question #21 of 120

Question ID: 1214806

The following table summarizes the results of a poll (hypothetically) taken of CFA charterholders and CFA candidates regarding the importance of a continuing education requirement after the CFA designation is obtained:

Group	In Favor of a Continuing Education Requirement	Against a Continuing Education Requirement
CFA charterholders	235	765
CFA candidates	180	820

Given the information that a member of the group is in favor of a continuing education requirement, what is the probability that she is a CFA candidate?

- A) 18%.
- B) 37%.
- C) 43%.

Explanation

There are 415 members of the group in favor of the continuing education requirement. 180 of these are candidates, so the probability is $180 / 415 = 43.37\%$. Note that this question is quite similar to those where we apply Bayes' Theorem. The priors here are the probabilities that a member of the group is a charterholder or is a candidate. Both of these are $1,000 / 2,000 = 50\%$. Given the

information that the member of the group favors continuing education, we can update the probability that that group member is a candidate to 43.37%.

For Further Reference:

(Study Session 2, Module 8.1, LOS 8.d)

CFA® Program Curriculum, Volume 1, page 462

Question #22 of 120

Question ID: 1214809

Which of the following statements about common probability distributions is *least accurate*?

- A) A probability distribution specifies the probabilities of the possible outcomes of a random variable.
- B) In a binomial probability distribution, each observation has only two possible outcomes that are mutually exclusive.
- C) **A normal distribution is a discrete symmetric probability distribution that is completely described by two parameters: its mean and variance.**

Explanation

A normal distribution is a *continuous* symmetric probability distribution.

For Further Reference:

(Study Session 3, Module 9.1, 9.2, LOS 9.a, 9.e, 9.i)

CFA® Program Curriculum, Volume 1, page 514

CFA® Program Curriculum, Volume 1, page 516

CFA® Program Curriculum, Volume 1, page 531

Question #23 of 120

Question ID: 1214808

For a binomial random variable with a 40% probability of success on each trial, the expected number of successes in 12 trials is *closest* to:

- A) **4.8.**
- B) 5.6.
- C) 7.2.

Explanation

A binomial random variable has an expected value or mean equal to np . Mean = $12(0.4) = 4.8$.

For Further Reference:

(Study Session 3, Module 9.1, LOS 9.f)

CFA® Program Curriculum, Volume 1, page 516

Question #24 of 120

Question ID: 1214810

An investor is considering investing in one of the following three portfolios:

Statistical Measures	Portfolio X	Portfolio Y	Portfolio Z
Expected annual return	12%	17%	22%
Standard deviation of return	14%	20%	25%

If the investor's minimum acceptable return is 5%, the optimal portfolio using Roy's safety-first criterion is:

- A) Portfolio X.
- B) Portfolio Y.
- C) **Portfolio Z.**

Explanation

$$\begin{aligned}\text{Portfolio X: SFRatio} &= \frac{12-5}{14} = 0.50 \\ \text{Portfolio Y: SFRatio} &= \frac{17-5}{20} = 0.60 \\ \text{Portfolio Z: SFRatio} &= \frac{22-5}{25} = 0.68\end{aligned}$$

According to the safety-first criterion, Portfolio Z, with the largest ratio (0.68), is the best alternative.

For Further Reference:

(Study Session 3, Module 9.3, LOS 9.m)

CFA® Program Curriculum, Volume 1, page 537

Question #25 of 120

Question ID: 1214815

An analyst is testing the hypothesis that the variance of monthly returns for Index L equals the variance of monthly returns for Index M based on samples of 50 monthly observations. The sample variance of Index L returns is 0.085, whereas the sample variance of Index M returns is 0.084. Assuming the samples are independent and the returns are normally distributed, which of the following represents the *most appropriate* test statistic?

- A) $\frac{\text{sample variance of Index L}}{\text{sample variance of Index M}}$
- B) $\frac{\text{variance of Index L} - \text{variance of Index M}}{\text{standard error of squared differences}}$
- C) $\frac{(\text{degrees of freedom}) \times (\text{Index L sample variance})}{\text{Index M sample variance}}$

Explanation

The appropriate test is an *F*-test, where the larger sample variance (Index L) is placed in the numerator.

For Further Reference:

(Study Session 3, Module 11.3, LOS 11.j)

CFA® Program Curriculum, Volume 1, page 636

Question #26 of 120

Question ID: 1214807

If the probability of event J multiplied by the probability of event K is not equal to the joint probability of events J and K, then events J and K are *most likely*:

- A) dependent events.
- B) independent events.
- C) mutually exclusive events.

Explanation

Events J and K are dependent. By the multiplication rule, joint probability $P(JK) = P(J|K) \times P(K)$, or $P(JK) = P(K|J) \times P(J)$. Events are independent if $P(J|K) = P(J)$ and $P(K|J) = P(K)$. This implies that for independent events, $P(JK) = P(J) \times P(K)$. If this condition is not met, events J and K are dependent. If events J and K are mutually exclusive, their joint probability is zero. The information given is consistent with this but not sufficient to conclude that this is the case.

For Further Reference:

(Study Session 2, Module 8.2, LOS 8.g)

CFA® Program Curriculum, Volume 1, page 467

Question #27 of 120

Question ID: 1214812

Alan Barnes, CFA, is interested in the expected return on the FTSE 100 stock index next quarter. He has data for the last five years and calculates the average return on the index over the last 20 quarters. This average return:

- A) is different from the parameter he is trying to estimate by the amount of the sampling error.
- B) overstates the return because he should divide by the square root of 20 when using a mean value.
- C) overstates the expected return because he should have used the geometric mean and not the simple average.

Explanation

The 20 quarters he has used are a sample of all the possible outcomes for the quarterly returns on the index. The difference between the true population parameter (mean index return) he is trying to estimate and the sample statistic he has calculated is called the sampling error. The arithmetic mean is the appropriate estimator of the next period's return.

For Further Reference:

(Study Session 2, Module 7.3, LOS 7.m and Study Session 3, Module 10.1, LOS 10.b)

CFA® Program Curriculum, Volume 1, page 570

Question #28 of 120

Question ID: 1214814

Which of the following statements regarding the significance level of a hypothesis test is *most accurate*?

- A) Given a significance level of 5%, a test will reject a true null hypothesis 5% of the time.**
- B)** If the significance level of a test is 5%, it will yield the correct decision about the null hypothesis 95% of the time.
- C)** If the significance level of a test is 95%, it will yield the correct decision about the null hypothesis 95% of the time.

Explanation

The significance level of a test is the probability that a true null hypothesis will be rejected by chance because the test statistic is from a sample and may take on a value that is outside the range of critical values because of sampling error. Choice B is incorrect because the probability of making a correct decision also must account for the probability of failing to reject a false null hypothesis.

For Further Reference:

(Study Session 3, Module 11.1, LOS 11.c)

CFA® Program Curriculum, Volume 1, page 613

Question #29 of 120

Question ID: 1214811

A researcher needs to choose a probability distribution for the price of an asset that is quite volatile in order to simulate returns outcomes. She has a program that will generate random variables from any of a variety of distributions. The *most appropriate* distribution for her to select to generate the asset price distribution is a:

- A)** normal distribution.
- B) lognormal distribution.**
- C)** Student's *t*-distribution.

Explanation

The lognormal distribution is most appropriate for modeling asset prices because the values cannot be less than zero and are not bounded on the upside. A binomial distribution allows only two possible outcomes over a period.

For Further Reference:

(Study Session 3, Module 9.3, LOS 9.n)

CFA® Program Curriculum, Volume 1, page 539

Question #30 of 120

Question ID: 1214813

Excerpts from statistical tables:

Student's t-Distribution

Level of Significance for Two-Tailed Test			
df	0.10	0.05	0.01
27	1.703	2.052	2.771
28	1.701	2.048	2.763
29	1.699	2.045	2.756

Cumulative z-Table

z	0.07	0.08	0.09
1.2	0.8980	0.8997	0.9015
1.3	0.9147	0.9162	0.9177
1.4	0.9292	0.9306	0.9319

An analyst has a sample of 28 observations of the mean weekly return of an index portfolio that includes 50 stocks. The weekly returns are approximately normally distributed, and the sample mean and sample variance are 1.2% and 0.00175. A 90% confidence interval for the population mean is *closest* to:

- A) -5.702% to 8.102%.
- B) -5.916% to 8.315%.
- C) -5.924% to 8.324%.

Explanation

Based on a t-distribution with 27 degrees of freedom, a 90% confidence interval for the mean is $1.2\% \pm 1.703 \times (0.00175)^{1/2} = -5.924\%$ to 8.324%.

For Further Reference:

(Study Session 3, Module 10.2, LOS 10.j)

CFA® Program Curriculum, Volume 1, page 581

Question #31 of 120

Question ID: 1214823

Questions 31 through 42 relate to Economics. (18 minutes)

If investors' expected future incomes increase and the demand for financial capital increases, other things equal:

- A) the equilibrium interest rate will rise.
- B) the equilibrium interest rate will fall.
- C) these two factors will have opposing effects on the equilibrium interest rate.

Explanation

Increases in expected future incomes will decrease savings, which will decrease the supply of financial capital and increase the equilibrium interest rate. If the demand for financial capital rises, interest rates also rise; so both changes tend to increase the equilibrium interest rate.

For Further Reference:

(Study Session 5, Module 16.1, LOS 16.d)

CFA® Program Curriculum, Volume 2, page 269

Question #32 of 120

Question ID: 1214824

Assume that the required reserve ratio is 20%, and banks currently have no excess reserves. If the Federal Reserve then buys \$100 million of Treasury bills from the banks, the money supply could potentially increase by:

- A) \$20 million.
- B) \$100 million.
- C) **\$500 million.**

Explanation

$$\text{Potential expansion multiplier} = \frac{1}{\text{required reserve ratio}} = \frac{1}{0.2} = 5$$

$$(100)(5) = 500$$

For Further Reference:

(Study Session 5, Module 16.1, LOS 16.c)

CFA® Program Curriculum, Volume 2, page 264

Question #33 of 120

Question ID: 1214821

If the number of employed remains constant, what effects will a decrease in the labor force have on the unemployment rate?

- A) Increase.
- B) **Decrease.**
- C) No effect.

Explanation

The unemployment rate is the number of unemployed divided by the labor force. Because the labor force is the sum of employed and unemployed, a decrease in the labor force with the number of employed held constant represents a decrease in the number of unemployed, and will decrease the unemployment rate.

For Further Reference:

(Study Session 4, Module 15.2, LOS 15.d)

CFA® Program Curriculum, Volume 2, page 219

Question #34 of 120

Question ID: 1214826

The country of Hokah can produce 35 units of cheese or 30 units of leather with one hour of labor. The country of Ymer can produce 20 units of cheese or 25 units of leather with one hour of labor. Which of the following statements is *most accurate*?

- A) **Ymer's opportunity cost of one unit of leather is 0.80 units of cheese.**
- B) Hokah's opportunity cost of one unit of cheese is 1.167 units of leather.
- C) Hokah has an absolute and a comparative advantage in both cheese and leather.

Explanation

The opportunity cost of one unit of leather for Ymer is $20 / 25 = 0.80$ units of cheese, and the opportunity cost of one unit of cheese is $25 / 20 = 1.25$ units of leather. The opportunity cost of one unit of cheese for Hokah is $30 / 35 = 0.86$ units of leather, and the opportunity

cost of one unit of leather is $35 / 30 = 1.167$ units of cheese. Hokah has an absolute advantage in both cheese and leather but has a comparative advantage only in cheese.

For Further Reference:

(Study Session 5, Module 17.1, LOS 17.c)

CFA® Program Curriculum, Volume 2, page 343

Question #35 of 120

Question ID: 1214820

Short-run disequilibrium is *most likely* to be associated with a decreasing price level if it results from a decrease in:

- A) both aggregate demand and short-run aggregate supply.
- B) aggregate demand, if short-run aggregate supply is unchanged.**
- C) short-run aggregate supply, if aggregate demand is unchanged.

Explanation

A decrease in aggregate demand, holding short-run aggregate supply constant, would decrease the price level. A decrease in short-run aggregate supply, holding aggregate demand constant, would increase the price level. If both aggregate demand and short-run aggregate supply decrease, the price level may increase or decrease.

For Further Reference:

(Study Session 4, Module 14.3, LOS 14.j)

SchweserNotes, Book 2 page 67

CFA® Program Curriculum, Volume 2, page 161

Question #36 of 120

Question ID: 1214818

If the government regulates a natural monopoly and enforces an average cost pricing, what are the effects on output quantity and price compared to an unregulated natural monopoly?

- A) Both are lower under average cost pricing.
- B) Both are higher under average cost pricing.
- C) One is higher and one is lower under average cost pricing.**

Explanation

Average cost pricing is meant to force a natural monopolist to reduce price to where the firm's average total cost intersects the market demand curve. This results in higher output and a lower price than would prevail for an unregulated natural monopoly.

For Further Reference:

(Study Session 4, Module 13.4, LOS 13.e)

CFA® Program Curriculum, Volume 2, page 64

Question #37 of 120

Question ID: 1214827

Assume that one year ago, the exchange rate between the Japanese yen and the euro was 100 JPY/EUR, and the exchange rate between the Japanese yen and the U.S. dollar was 80 JPY/USD. Current exchange rates are 104.2 JPY/EUR and 76.6 JPY/USD. Which of the following statements is *most accurate*?

- A) The USD has depreciated relative to the EUR.**
- B) The JPY has depreciated 4.2% relative to the EUR.**
- C) The current U.S. dollar to euro exchange rate is approximately 1.25 USD/EUR.**

Explanation

We can calculate the current USD/EUR cross rate as $104.2 / 76.6 = 1.3603$ USD/EUR. The original USD/EUR cross rate was $100 / 80 = 1.2500$ USD/EUR. Thus, the USD has depreciated relative to the EUR. While it is correct to say that the EUR has appreciated 4.2% relative to the JPY ($104.2 / 100 - 1 = 4.2\%$), it is not correct to say that the JPY has depreciated by the same percentage. To calculate the percentage change in the JPY relative to the EUR, we need to invert the quotes. One year ago, the quote was 0.0100 EUR/JPY and now the quote is 0.0096 EUR/JPY. $(0.0096 / 0.0100 - 1) = 0.0403$ or 4.0% depreciation in the JPY relative to the EUR.

For Further Reference:

(Study Session 5, Module 18.1, LOS 18.c, 18.d)

CFA® Program Curriculum, Volume 2, page 414

CFA® Program Curriculum, Volume 2, page 417

Question #38 of 120

Question ID: 1214817

Oligopolists have an incentive to cheat on collusive agreements in order to:

- A) avoid competitive practices.**
- B) increase their individual share of the joint profit.**
- C) restrict output and put upward pressure on price.**

Explanation

Colluding restricts output and puts upward pressure on price, but cheating actually increases output and ultimately, if enough cheating occurs, puts downward pressure on the price. Colluders cheat to increase their share of the profits.

For Further Reference:

(Study Session 4, Module 13.3, LOS 13.d)

CFA® Program Curriculum, Volume 2, page 64

Question #39 of 120

Question ID: 1214822

If the inflation rate was -4% in 20X1, -2% in 20X2, and -1% in 20X3, the economy is *most accurately* described as experiencing:

- A) deflation.**
- B) contraction.**
- C) disinflation.**

Explanation

If the inflation rate is negative, an economy is said to be experiencing deflation. Deflation is often, but not necessarily, associated with business cycle contractions.

For Further Reference:

(Study Session 4, Module 15.2, LOS 15.e)

CFA® Program Curriculum, Volume 2, page 223

Question #40 of 120

Question ID: 1214819

With respect to the IS-LM model, in an LM curve the real interest rate is:

- A) positively related to real income, holding the real money supply constant.**
- B) held constant, resulting in excess savings being positively related to real income.
- C) negatively related to real income, holding the marginal propensity to save constant.

Explanation

The LM curve illustrates a positive relationship between real income and the real interest rate, holding the real money supply constant. The IS curve illustrates a negative relationship between real income and the real interest rate, holding the marginal propensity to save constant.

For Further Reference:

(Study Session 4, Module 14.2, LOS 14.f)

CFA® Program Curriculum, Volume 2, page 135

Question #41 of 120

Question ID: 1214816

An individual sees her income rise from \$80,000 to \$88,000, and along with it, her consumption of Good X has decreased from eight dozen packages per year to six dozen packages per year. Good X should be classified as:

- A) a normal good.**
- B) a Veblen good.
- C) an inferior good.**

Explanation

An inferior good is one that experiences a decline in demand when income rises.

For Further Reference:

(Study Session 4, Module 12.2, LOS 12.c)

CFA® Program Curriculum, Volume 2, page 19

Question #42 of 120

Question ID: 1214825

The Keynesian view suggests that the government can reduce aggregate demand by using:

- A) restrictive fiscal policy to shift the government budget toward a surplus (or smaller deficit).**
- B) restrictive fiscal policy to shift the government budget toward a deficit (or a smaller surplus).
- C) expansionary fiscal policy to shift the government budget toward a surplus (or a smaller deficit).

Explanation

According to the Keynesians, policymakers can use the budget to diminish aggregate demand through restrictive fiscal policy. Reducing government expenditures and/or increasing tax rates should lead to a decline in the expected size of the budget deficit or an increase in the budget surplus.

For Further Reference:

(Study Session 5, Module 16.3, LOS 16.s)

CFA® Program Curriculum, Volume 2, page 315

Question #43 of 120

Question ID: 1214835

Questions 43 through 60 relate to Financial Reporting and Analysis. (27 minutes)

Selected year-end values for Able, Inc., are presented in the following tables:

Balance Sheet Items	December 31, 20X2	December 31, 20X1
Accounts receivable	\$19 million	\$21 million
Trade payables	\$14 million	\$16 million
Inventory	\$31 million	\$28 million

Income Statement Items	December 31, 20X2
Net income	\$42 million
Loss on sale	\$3 million
Depreciation	\$2 million
Dividends paid	\$1 million

Based only on the data in the tables, 20X2 cash flow for operations for Able, Inc., is:

- A) \$43 million.**
- B) \$44 million.**
- C) \$45 million.

Explanation

Starting with net income of 42, add 2 for the decrease in receivables (a source of cash), subtract 2 for the decrease in payables (a use of cash), subtract 3 for the increase in inventory (a use of cash), add 3 for the loss (a non-CFO item), and add 2 for depreciation (a noncash item), resulting in CFO = \$44 million. Note that the balance sheet items are presented from right to left.

For Further Reference:

Question #44 of 120

Question ID: 1214831

Which of the following items affects owners' equity but is not included as a component of net income?

- A) Depreciation.
- B) Dividends received on shares of another company classified as available for sale.
- C) Foreign currency translation gains and losses.**

Explanation

Foreign currency translation gains and losses are not reported on the income statement as a component of net income, but affect owners' equity because they are included as other comprehensive income. The other items are included on the income statement so they affect both net income and owners' equity.

For Further Reference:

(Study Session 7, Module 21.5, LOS 21.I)

CFA® Program Curriculum, Volume 3, page 109

Question #45 of 120

Question ID: 1214839

Gordon, Inc., reports using the LIFO cost method. In notes to its financial statements, Gordon discloses inventory of 500 units in 20X1 and 510 units in 20X2 and states a LIFO reserve of \$50,000 in 20X1 and \$48,000 in 20X2. These inventory disclosures *most likely* reflect:

- A) decreasing demand.
- B) decreasing prices.**
- C) a LIFO liquidation.

Explanation

Decreasing prices can cause the LIFO reserve to decrease even when the inventory quantity is stable or increasing. A LIFO liquidation is a decrease in the quantity of inventory. To look for an indication of decreasing demand, an analyst would compare the growth rates of inventories and sales.

For Further Reference:

(Study Session 8, Modules 25.2, 25.3, 25.4, LOS 25.d, 25.e, 25.j)

CFA® Program Curriculum, Volume 3, page 320

CFA® Program Curriculum, Volume 3, page 323

CFA® Program Curriculum, Volume 3, page 339

Question #46 of 120

Question ID: 1214837

Which of the following statements about financial ratios is *most accurate*?

- A) Any firm with a high net profit margin will have a high gross profit margin and vice versa.
- B) A company with a high debt-to-equity ratio will have a return on assets that is greater than its return on equity.
- C) A company that has an inventory turnover of 6 times, a receivables turnover of 9 times, and a payables turnover of 12 times will have a cash conversion cycle of approximately 71 days.

Explanation

The cash conversion cycle is $(365 / 6) + (365 / 9) - (365 / 12) = 60.8 + 40.6 - 30.4 = 71$ days. ROA is less than ROE when net income is positive and debt is present. The fact that a company has a high gross profit margin does not necessarily mean it will have a high net profit margin. A company with a high gross margin may have a low (or negative) net margin if its operating expenses are high.

For Further Reference:

(Study Session 7, Module 24.2, LOS 24.b)

CFA® Program Curriculum, Volume 3, page 256

Question #47 of 120

Question ID: 1214836

An analyst gathered the following information about a company:

Cash flow from operations	\$800
Purchase of plant and equipment	40
Sale of land	30
Interest expense	80
Depreciation and amortization	100

The company has a tax rate of 35% and prepares its financial statements under U.S. GAAP.

The company's free cash flow to the firm (FCFF) is *closest* to:

- A) \$840.
- B) \$870.
- C) \$940.

Explanation

FCFF = Cash flow from operations + interest expense net of tax – net capital expenditures

$$\text{FCFF} = \$800 + 80(1 - 0.35) - 40 + 30 = \$842$$

Depreciation and amortization do not have to be added when calculating FCFF from CFO. They are added when calculating FCFF from net income.

For Further Reference:

(Study Session 7, Module 23.4, LOS 23.i)

CFA® Program Curriculum, Volume 3, page 222

Question #48 of 120

Question ID: 1214832

For which of the following balance sheet items is a change in market value *most likely* to affect net income?

- A) Debt securities issued by the firm.
- B) Equity securities purchased by the firm.
- C) Debt securities that the firm intends to hold until maturity.

Explanation

Listed equity securities are measured at fair value through profit and loss, unless the firm chooses at the time of purchase to measure them at fair value through other comprehensive income. Debt securities issued by the firm, and debt securities that the firm intends to hold until maturity, are both reported at amortized cost unless the firm chooses to report them at market value.

For Further Reference:

(Study Session 7, Module 22.6, LOS 22.e)

CFA® Program Curriculum, Volume 3, page 133

Question #49 of 120

Question ID: 1214828

Gus Davy, CFA, is reviewing an industry that has been experiencing rising prices as well as unit volume growth. Davy's investment criteria include selecting companies generating the highest profit margins. If Davy does not adjust companies' financial statements for their inventory cost assumptions, he is *most likely* to select companies that use:

- A) FIFO.
- B) LIFO.
- C) weighted average cost.

Explanation

The FIFO method recognizes the oldest costs in the cost of goods sold. With rising prices, COGS will be lower and net income will be higher using FIFO as compared to the LIFO or average cost methods. Higher net income relative to sales (which are not affected by the inventory cost method) means higher profit margins.

For Further Reference:

(Study Session 7, Module 21.3, LOS 21.d)

CFA® Program Curriculum, Volume 3, page 84

Question #50 of 120

Question ID: 1214844

A company using LIFO reports the following:

- Cost of goods sold was \$27,000.
- Beginning inventory was \$6,500, and ending inventory was \$6,200.
- The beginning LIFO reserve was \$1,200.
- The ending LIFO reserve was \$1,400.

The *best* estimate of the company's cost of goods sold on a FIFO basis would be:

- A) \$21,300.
- B) \$26,800.**
- C) \$27,500.

Explanation

COGS FIFO = COGS LIFO – (ending LIFO reserve – beginning LIFO reserve)

$$\text{COGS FIFO} = 27,000 - (1,400 - 1,200) = \$26,800.$$

For Further Reference:

(Study Session 8, Module 25.3, LOS 25.f and Study Session 9, Module 30.2, LOS 30.e)

CFA® Program Curriculum, Volume 3, page 323

CFA® Program Curriculum, Volume 3, page 627

Question #51 of 120

Question ID: 1214830

An analyst gathered the following data about a company:

- 1,000,000 shares of common are outstanding at the beginning of the year.
- 10,000 6% convertible bonds (conversion ratio is 20 to 1) were issued at par June 30 of this year.
- The company has 100,000 warrants outstanding all year with an exercise price of \$25 per share.
- The average stock price for the period is \$20, and the ending stock price is \$30.

If the convertible bonds are considered dilutive, the number of shares of common stock that the analyst should use to calculate diluted earnings per share is:

- A) 1,000,000.
- B) 1,100,000.**
- C) 1,266,667.

Explanation

When the capital structure contains options or warrants, the treasury stock method uses the average price. In this situation, the warrants are antidilutive because the exercise price of the warrant (\$25) is higher than the market price of the stock (\$20). Thus, warrants are excluded. Otherwise, common shares would be reduced.

$$\text{original shares of common stock} = 1,000,000 \left(\frac{12}{12} \right)$$

Then add the impact of the bond conversion:

$$(10,000)(20) \times (6/12) = 100,000$$

Thus, the adjusted denominator for fully diluted EPS is:

$$1,000,000 + 100,000 = 1,100,000$$

For Further Reference:

(Study Session 7, Module 21.4, LOS 21.g, 21.h)

Question #52 of 120

Question ID: 1214840

Degen, Inc., owns a trademark which it originally valued at €15 million on its balance sheet but currently values at €10 million. In the country where Degen is incorporated, trademarks are protected by law for as long as their owner remains a going concern. Degen has *most likely*:

- A) developed its trademark at a cost of €15 million.
- B) recorded amortization expense of €5 million on its trademark.
- C) recognized €5 million of impairment charges on its trademark.**

Explanation

The trademark is an intangible asset with an indefinite life, and its cost is not amortized. The decrease in the trademark's balance sheet value must be the result of impairment. For the intangible asset to appear on the balance sheet, Degen must have purchased the trademark. If Degen had developed the trademark internally, it would have expensed the cost rather than capitalizing it to the balance sheet.

For Further Reference:

(Study Session 8, Modules 26.1, 26.3, LOS 26.a, 26.b, 26.i)

CFA® Program Curriculum, Volume 3, page 379

SchweserNotes, Book 3 page 181

CFA® Program Curriculum, Volume 3, page 382

CFA® Program Curriculum, Volume 3, page 411

Question #53 of 120

Question ID: 1214838

Which of the following ratios is a component of the original (three-part) DuPont equation?

- A) Asset turnover.**
- B) Gross profit margin.
- C) Debt-to-equity ratio.

Explanation

The three-part DuPont approach is as follows: net profit margin × asset turnover × leverage ratio, where the leverage ratio is assets-to-equity.

For Further Reference:

(Study Session 7, Module 24.4, LOS 24.d)

CFA® Program Curriculum, Volume 3, page 279

Question #54 of 120

Question ID: 1214829

Harding Corp. has a permanently impaired asset. The difference between its carrying value and the present value of its expected cash flows should be written down immediately and:

- A) reported as an operating loss.
- B) charged directly against retained earnings.
- C) reported as a non-operating loss in other comprehensive income.

Explanation

Impairment writedowns are reported losses "above the line" and are included in income from continuing operations.

For Further Reference:

(Study Session 7, Module 21.3, LOS 21.e)

CFA® Program Curriculum, Volume 3, page 93

Question #55 of 120

Question ID: 1214842

If market interest rates have changed materially since a firm issued a bond, and the firm uses the effective interest rate method, how is a change in the market value of the firm's debt *most likely* to be reported in the firm's financial statements?

- A) The gain or loss in market value must be calculated and disclosed in the footnotes to the financial statements.
- B) **Net income and equity are unaffected, but the change may be discussed in management's commentary.**
- C) Net income is unaffected, but the change in market value is recorded in other comprehensive income.

Explanation

Material changes in the firm's cost of debt capital should be included in the Management Discussion and Analysis section of the financial statements. If the firm does not use fair value reporting of debt obligations, net income and shareholders' equity are not affected by changes in the market value of the firm's debt, and disclosing its gain or loss in market value is not required.

For Further Reference:

(Study Session 8, Module 28.3, LOS 28.e)

CFA® Program Curriculum, Volume 3, page 506

Question #56 of 120

Question ID: 1214841

Which of the following statements regarding deferred taxes is *least* accurate?

- A) A permanent difference is a difference between taxable income and pretax income that will not reverse.
- B) A deferred tax asset is created when a temporary difference results in taxable income that exceeds pretax income.

C) Deferred tax assets and liabilities are not adjusted for changes in tax rates.

Explanation

Deferred tax assets and liabilities are adjusted for changes in expected tax rates under the liability method.

For Further Reference:

(Study Session 8, Module 27.4, LOS 27.e)

CFA® Program Curriculum, Volume 3, page 461

Question #57 of 120

Question ID: 1214833

The direct method of reporting operating cash flow:

- A) is preferred by analysts and commercial lenders.**
- B) is the most frequently used method under both U.S. GAAP and IFRS.**
- C) shows the reasons for differences between net income and operating cash flows.**

Explanation

The direct method is preferred by analysts and commercial lenders because it gives information about specific types of operating cash flows that is useful in determining a company's future financing needs and ability to repay debt. The indirect method shows the reasons for differences between net income and operating cash flows.

For Further Reference:

(Study Session 7, Module 23.1, LOS 23.e)

CFA® Program Curriculum, Volume 3, page 198

Question #58 of 120

Question ID: 1214843

Jordan Loney, CFA, issues a "sell" recommendation on Sullivan Company because she believes its financial reporting quality is low. Loney writes, "Rapid turnover of key employees in its information systems and accounting units have made Sullivan's internal monitoring controls ineffective." Which condition that may lead to low-quality financial reporting has Loney detected at Sullivan?

- A) Opportunity.**
- B) Motivation.**
- C) Rationalization.**

Explanation

Ineffective internal controls over accounting may provide opportunities for low-quality financial reporting.

For Further Reference:

(Study Session 9, Module 29.1, LOS 29.e)

CFA® Program Curriculum, Volume 3, page 558

Question #59 of 120

Question ID: 1214877

At the beginning of the year, a firm securitizes half of its accounts receivable and uses the proceeds to pay down principal on a long-term bank loan. These transactions will:

- A) increase the firm's current ratio.
- B) decrease the firm's debt-to-equity ratio.**
- C) decrease the firm's interest coverage ratio.

Explanation

Securitizing accounts receivable means the firm sells them to a special purpose entity that will issue asset-backed securities. This transaction will remove these accounts receivable from the firm's balance sheet. Using the proceeds to retire long-term debt will decrease liabilities by the same amount. Shareholders' equity is not affected. Therefore, the debt-to-equity ratio will decrease. The current ratio will decrease because accounts receivable (a current asset) decrease while current liabilities remain unchanged (or decrease only by any current portion of the debt paid down). The interest coverage ratio will increase because paying down long-term debt will decrease interest expense.

For Further Reference:

(Study Session 7, Modules 22.7, 24.3, LOS 22.h, 24.b and Study Session 14, Module 45.1, LOS 45.b)

CFA® Program Curriculum, Volume 3, page 169

CFA® Program Curriculum, Volume 3, page 256

CFA® Program Curriculum: Volume 5, page 181

Question #60 of 120

Question ID: 1214834

Interest paid is reported as an operating cash outflow under:

- A) U.S. GAAP, but may be reported as a financing cash flow under IFRS.**
- B) IFRS, but may be reported as an investing cash flow under U.S. GAAP.
- C) U.S. GAAP, but must be reported as either an investing or financing cash flow under IFRS.

Explanation

Interest paid is an operating cash flow under U.S. GAAP but may be reported as either an operating or financing cash flow under IFRS.

For Further Reference:

(Study Session 7, Module 23.1, LOS 23.d)

CFA® Program Curriculum, Volume 3, page 188

Question #61 of 120

Question ID: 1214853

Questions 61 through 72 relate to Corporate Finance. (18 minutes)

Business risk is *best* described as resulting from the combined effects of a firm's:

- A) financial risk and sales risk.**

B) sales risk and operating risk.

C) operating risk and financial risk.

Explanation

Business risk is the combination of sales risk, which is the variability of a firm's sales, and operating risk, which is the additional variability in operating earnings (EBIT) caused by fixed operating costs.

For Further Reference:

(Study Session 11, Module 34.1, LOS 34.a)

CFA® Program Curriculum, Volume 4, page 126

Question #62 of 120

Question ID: 1214845

Which of the following relationships among a company's stakeholders is *most accurately* described as a principal-agent relationship?
The relationship between a company's managers and its:

A) creditors.

B) regulators.

C) shareholders.

Explanation

The relationship between shareholders and managers of a company is a principal-agent relationship because the shareholders (the principals) hire managers (the agents) to act in the best interests of the shareholders.

For Further Reference:

(Study Session 10, Module 31.1, LOS 31.c)

CFA® Program Curriculum, Volume 4, page 11

Question #63 of 120

Question ID: 1214850

The amount of a company's optimal capital budget is *most accurately* determined by the point on the company's investment opportunity schedule:

A) where the amount of new capital raised is at its minimum.

B) where it intersects the company's marginal cost of capital curve.

C) where the expected return on the next potential project is at its maximum.

Explanation

The investment opportunity schedule is a downward sloping curve of the internal rates of return (expected returns) of potential projects ranked from highest to lowest. This curve intersects the company's upward sloping marginal cost of capital curve at an amount of capital where the marginal project's IRR just equals the firm's cost of capital. The firm should accept projects with IRRs that exceed the marginal cost of capital (lie to the left of the intersection) and reject projects with IRRs less than the marginal cost of capital (lie to the right of the intersection).

For Further Reference:

Question #64 of 120

Question ID: 1214855

A company is *most likely* faced with a drag on liquidity if its:

- A) weighted average collection period increases from 42 days to 46 days.**
- B)** largest vendor changes its invoice terms from "3/10 net 30" to "3/10 net 60."
- C)** inventory turnover was below the industry average last year and is above the industry average this year.

Explanation

An increase in the weighted average collection period indicates that customers are taking longer to pay their outstanding accounts. This represents a drag on the company's liquidity. A vendor that changes its payment terms from "net 30" to "net 60" is allowing the company 60 days to pay instead of 30. This extension of trade credit is a source of liquidity for the company. An inventory turnover ratio that is increasing relative to the industry average is a sign of good inventory management, which can also be a source of liquidity for a company.

For Further Reference:

(Study Session 11, Module 35.1, LOS 35.a, 35.f)

CFA® Program Curriculum, Volume 4, page 157

CFA® Program Curriculum, Volume 4, page 175

Question #65 of 120

Question ID: 1214846

A portfolio manager who tracks the industry weights of a benchmark index, but within each industry selects stocks with favorable ESG factors, is said to be engaging in:

- A)** active ownership.
- B)** thematic investing.
- C) best-in-class investing.**

Explanation

With a best-in-class investing strategy, a portfolio manager selects the stocks within each industry or sector that have the most favorable ESG characteristics, without excluding any industries. Active ownership refers to using share ownership as a way to promote ESG-related goals, through proxy voting and access to company management. Thematic investing strategies are based on a single ESG factor.

For Further Reference:

(Study Session 10, Module 31.1, LOS 31.k)

CFA® Program Curriculum, Volume 4, page 40

Question #66 of 120

Question ID: 1214851

Faye Harlan, CFA, is estimating the cost of common equity for Cyrene Corporation. She prepares the following data for Cyrene:

- Price per share = \$50.
- Expected dividend per share = \$3.
- Expected retention ratio = 30%.
- Expected return on equity = 20%.
- Beta = 0.89.
- Yield to maturity on outstanding debt = 10%.
- The expected market rate of return is 12% and the risk-free rate is 3%.

Based on these data, Harlan determines that Cyrene's cost of common equity is 14%. Harlan *most likely* arrived at this estimate by using the:

- A) dividend discount model approach.**
- B) capital asset pricing model approach.**
- C) bond yield plus risk premium approach.**

Explanation

Using the CAPM approach, the estimated cost of common equity = $3\% + 0.89(12\% - 3\%) = 11\%$. Using the dividend discount model approach, the growth rate = $(0.3)(0.2) = 6\%$ and the estimated cost of common equity = $\$3 / \$50 + 6\% = 12\%$. To get a cost of common equity of 14%, Harlan most likely added a risk premium to Cyrene's bond yield.

For Further Reference:

(Study Session 10, Module 33.1, LOS 33.h)

CFA® Program Curriculum, Volume 4, page 88

Question #67 of 120

Question ID: 1214856

Quixote Co. and Sisyphus Co., two similar-sized competitors, have had stable operating cycles of 180 days and cash conversion cycles of 140 days over the past several years. Sisyphus' operating and cash conversion cycles remained at these levels in the most recent year, but Quixote's cash conversion cycle contracted to 120 days while its operating cycle remained at 180 days. Relative to Sisyphus, Quixote has *most likely* begun:

- A) taking more time to pay its suppliers.**
- B) operating with less inventory on hand.**
- C) offering easier credit terms to its customers.**

Explanation

The cash conversion cycle is equal to the operating cycle minus the number of days of payables. If Quixote is extending the time it takes to pay its suppliers, its number of days of payables will increase, and its cash conversion cycle will decrease. Its operating cycle (days of inventory plus days of receivables) is unaffected by the increase in days of payables. Changes in inventory or receivables management would affect both the operating cycle and the cash conversion cycle.

For Further Reference:

(Study Session 11, Module 35.1, LOS 35.c)

CFA® Program Curriculum, Volume 4, page 159

Question #68 of 120

Question ID: 1214849

A company prepares a chart with the net present value (NPV) profiles for two mutually exclusive projects with equal lives of five years. Project Jones and Project Smith have the same initial cash outflow and total undiscounted cash inflows, but 75% of the cash inflows for Project Jones occur in years 1 and 2, while 75% of the cash inflows for Project Smith occur in years 4 and 5. Which of the following statements is *most accurate* regarding these projects?

- A) Project Smith has a higher internal rate of return than Project Jones.
- B) There is a range of discount rates in which the optimal decision is to reject both projects.**
- C) There is a range of discount rates in which the company should choose Project Jones and a range in which it should choose Project Smith.

Explanation

If the total undiscounted cash flows from two projects are equal, their NPV profiles intersect the vertical axis at the same value. The NPV profile will have a steeper slope for Project Smith, which has more of its cash inflows occurring later in its life, and therefore the IRR of Project Smith (its intersection with the horizontal axis) must be less than the IRR of Project Jones. The NPV for Project Jones will be greater at any rate of discount, and Project Jones will be preferred over the entire range. However, if the discount rate applied to the cash flows is greater than the IRR of Project Jones, both projects will have negative NPVs and the company should reject both of them.

For Further Reference:

(Study Session 10, Module 32.2, LOS 32.e)

CFA® Program Curriculum, Volume 4, page 58

Question #69 of 120

Question ID: 1214848

Two of the typical steps in the capital budgeting process are:

- A) analyzing project proposals and raising additional capital.
- B) raising additional capital and creating the firm-wide capital budget.
- C) analyzing project proposals and creating the firm-wide capital budget.**

Explanation

The four typical steps in the capital budgeting process are generating ideas, analyzing project proposals, creating the firm-wide capital budget, and monitoring decisions and conducting a post-audit. Raising additional capital is not one of the typical steps in the capital budgeting process.

For Further Reference:

(Study Session 10, Module 32.1, LOS 32.a)

CFA® Program Curriculum, Volume 4, page 48

Question #70 of 120

Question ID: 1214847

Which of the following board committees is *most likely* to be required by regulations?

- A) Audit committee.**
- B) Nominations committee.**
- C) Compensation committee.**

Explanation

While board committees may vary based on the size and complexity of companies, regulations often require an audit committee to be appointed.

For Further Reference:

(Study Session 10, Module 31.1, LOS 31.f)

CFA® Program Curriculum, Volume 4, page 21

Question #71 of 120

Question ID: 1214854

Under what conditions will the percentage change in a company's earnings per share be larger than the percentage change in its earnings before interest and taxes (EBIT)? The company's degree of:

- A) operating leverage is positive.**
- B) total leverage is greater than one.**
- C) financial leverage is greater than one.**

Explanation

Earnings per share will increase by a greater percentage than EBIT if a company has financial leverage (i.e., a degree of financial leverage greater than one).

For Further Reference:

(Study Session 11, Module 34.1, LOS 34.b, 34.c)

CFA® Program Curriculum, Volume 4, page 128

CFA® Program Curriculum, Volume 4, page 137

Question #72 of 120

Question ID: 1214852

A new project will require a company to issue both debt and common equity. In estimating the project's net present value (NPV), an analyst increases the initial outlay for this project by the flotation cost of the equity securities. Has the analyst treated flotation costs correctly?

- A) Yes.**
- B) No, because flotation costs are a sunk cost.**
- C) No, because the cost of capital should be adjusted for flotation costs.**

Explanation

Adding flotation costs attributable to a project to its initial cash outlay is the correct treatment.

For Further Reference:

Question #73 of 120

Question ID: 1214864

Questions 73 through 86 relate to Equity Investments. (21 minutes)

An analyst classifies Mettler, Inc., an operator of retail grocery stores, in the same industry group as Powell Corporation, a manufacturer of industrial machinery. This analyst's classification system is *most likely* based on:

- A) statistical methods.**
- B) products and services.**
- C) sensitivity to the business cycle.**

Explanation

This grouping is most likely the result of a cluster analysis and may have arisen by chance. The two firms clearly offer different products and services. An industrial machinery manufacturer is likely to be a cyclical firm, while a grocery retailer is likely to be a non-cyclical firm.

For Further Reference:

(Study Session 13, Module 40.1, LOS 40.b, 40.c)

CFA® Program Curriculum, Volume 4, page 389

CFA® Program Curriculum, Volume 4, page 390

Question #74 of 120

Question ID: 1214859

To ensure the continuity of a value-weighted index when one of the stocks in the index is split:

- A) no adjustment is necessary.**
- B) only the denominator must be adjusted for the split.**
- C) both the numerator and the denominator must be adjusted for the split.**

Explanation

Value-weighted indexes do not need to be adjusted for stock splits because the market capitalization of the company remains the same.

For Further Reference:

(Study Session 12, Module 37.1, LOS 37.e)

CFA® Program Curriculum, Volume 4, page 285

Question #75 of 120

Question ID: 1214870

Robert Higgins is estimating the price-earnings (P/E) ratio that will be appropriate for an index at the end of next year. He has estimated that:

- Expected annual dividends will increase by 10% compared to this year.

- Expected earnings per share will increase by 10% compared to this year.
- The expected growth rate of dividends will be the same as the current estimate of 5%.
- The required rate of return will rise from 8% to 11%.

Compared to the current P/E, the end-of-the-year P/E will be:

- A) 50% lower.**
- B) 2% higher.**
- C) 10% higher.**

Explanation

The numerator of the formula for the P/E is the payout ratio, which is unchanged (both expected earnings and dividends increase by the same percentage). The denominator ($k - g$) doubles from 3% to 6%, which will decrease the P/E by half.

For Further Reference:

(Study Session 13, Module 41.3, LOS 41.j)

SchweserNotes, Book 4 page 215

CFA® Program Curriculum, Volume 4, page 468

Question #76 of 120

Question ID: 1214867

An analyst gathered the following data about a stock:

- The stock paid a \$1 dividend last year.
- Next year's dividend is projected to be 10% higher.
- The stock is projected to sell for \$25 at the end of the year.
- The risk-free rate of interest is 8%.
- The expected return on the market is 13%.
- The stock's beta is 1.2.

The value of the stock today is *closest* to:

- A) \$19.45.**
- B) \$22.89.**
- C) \$26.74.**

Explanation

$$k = RFR + \beta(R_M - RFR) = 0.08 + 1.2(0.13 - 0.08) = 0.08 + 0.06 = 0.14$$

$$P_0 = \frac{P_1 + D_1}{1 + k} = \frac{25 + 1.10}{1.14} = \$22.89$$

For Further Reference:

(Study Session 13, Module 41.2, LOS 41.g)

CFA® Program Curriculum, Volume 4, page 459

Question #77 of 120

Question ID: 1214857

A contract that requires one party to pay \$100,000 each quarter to another company that will make a variable quarterly payment based on the market value of an equities portfolio is referred to as:

- A) a swap.**
- B) an index option.**
- C) portfolio insurance.**

Explanation

In an equity swap, one party pays a fixed amount periodically in return for a payment that is calculated using a specific amount and the return on a stock or portfolio of stocks.

For Further Reference:

(Study Session 12, Module 36.1, LOS 36.c)

CFA® Program Curriculum, Volume 4, page 218

Question #78 of 120

Question ID: 1214866

Visser, Inc. is an unprofitable fishing enterprise. Visser rents most of its boats and equipment but owns valuable transferable fishing quotas. If a competitor is interested in acquiring Visser, the *most appropriate* equity valuation model to use is:

- A) an asset-based valuation model.**
- B) an earnings multiplier model.**
- C) a Gordon growth model.**

Explanation

Asset-based models are often used to value natural resources companies and companies that are being liquidated. Because Visser is unprofitable, an earnings multiplier model or a dividend discount model such as the Gordon growth model may not produce a meaningful value for the firm.

For Further Reference:

(Study Session 13, Module 41.1, LOS 41.b)

CFA® Program Curriculum, Volume 4, page 448

Question #79 of 120

Question ID: 1214868

A firm has a constant growth rate of 7% and just paid a dividend of \$6.25. If the required rate of return is 12%, what will the stock sell for two years from now based on the dividend discount model?

- A) \$133.75.**
- B) \$149.80.**
- C) \$153.13.**

Explanation

$$\text{value @ } t = 2 = \frac{D_3}{k-g} = \frac{D_0(1+g)^3}{k-g} = \frac{\$6.25(1.07)^3}{0.12-0.07} = \$153.13$$

For Further Reference:

(Study Session 13, Module 41.2, LOS 41.g)

CFA® Program Curriculum, Volume 4, page 459

Question #80 of 120

Question ID: 1214863

In a transaction referred to as a management buyout (MBO):

- A) management sells its shares to an investor group attempting to gain control of a company.
- B) management buys a controlling interest in a public company to gain control of the board of directors.
- C) an investor group that includes management buys all the shares of a company and they no longer trade on an exchange.**

Explanation

Management buyout refers to a situation where an investor group that includes the firm's key management purchases all the outstanding shares (not just a controlling interest) of a public company in order to take it private. Once this is done, the shares are no longer registered for public trading and, as a result, are no longer traded on exchanges or in other public markets.

For Further Reference:

(Study Session 13, Module 39.1, LOS 39.c)

CFA® Program Curriculum, Volume 4, page 362

Question #81 of 120

Question ID: 1214865

Archer Products is in an industry that has experienced low levels of price competition but recently excess capacity has led to aggressive price cutting. An analyst would be *least likely* to describe Archer's industry as:

- A) concentrated and with high barriers to exit.
- B) in the shakeout stage with low concentration.
- C) in the maturity stage with high barriers to entry.**

Explanation

The maturity phase of an industry life cycle is not typically characterized by excess capacity or price competition. The conditions described may be consistent with a concentrated industry (typically characterized by low levels of price competition, since firms in the industry seek to avoid it) that is moving into the decline phase of an industry life cycle (excess capacity leads to aggressive price cutting, especially when barriers to exit are high). These conditions could also be consistent with an industry that is moving into the shakeout phase (characterized by developing excess capacity and intense price competition) following the growth phase (during which price competition is low).

For Further Reference:

(Study Session 13, Module 40.2, LOS 40.h)

Question #82 of 120

Question ID: 1214869

Assume the Wansch Corporation is expected to pay a dividend of \$2.25 per share this year. Sales and profit for Wansch are forecasted to grow at a rate of 20% for two years after that, then grow at 5% per year forever. Dividend and sales growth are expected to be equal. If Wansch's shareholders require a 15% return, the per-share value of Wansch's common stock based on the dividend discount model is *closest* to:

- A) \$22.75.
- B) \$26.00.
- C) \$28.50.

Explanation

$$P_2 = \frac{D_3}{k - g} = \frac{2.25(1.2)^2}{0.15 - 0.05} = 32.40$$

$$P_0 = \frac{\$2.25}{1.15} + \frac{2.70}{(1.15)^2} + \frac{32.40}{(1.15)^2} = \$28.50$$

For Further Reference:

(Study Session 13, Module 41.2, LOS 41.g)

CFA® Program Curriculum, Volume 4, page 459

Question #83 of 120

Question ID: 1214861

Under which of the following conditions are market values of securities *most likely* to be persistently greater than their intrinsic values?

- A) Short selling is restricted.
- B) Transactions costs are high.
- C) Arbitrage trading is restricted.

Explanation

Restrictions on short sales remove some of the selling pressure on overvalued securities that would otherwise drive their market values down toward their intrinsic values in an efficient market. Market inefficiency caused by high transactions costs or restrictions on arbitrage trading may allow market values that are different from intrinsic values to persist both for overvalued and undervalued securities.

For Further Reference:

(Study Session 12, Module 38.1, LOS 38.b, 38.c)

CFA® Program Curriculum, Volume 4, page 321

CFA® Program Curriculum, Volume 4, page 322

Question #84 of 120

Question ID: 1214858

Moore Company stock is currently trading at \$40 per share. An investor attempting to protect against losses of more than 10% on a short position in Moore should place a:

- A) stop buy order at \$44.**
- B) stop sell order at \$36.
- C) limit buy order at \$44.

Explanation

The investor is short the stock and will experience losses if the stock price increases. A stop buy order at \$44 will execute if the stock price rises to \$44 or higher. A limit buy order at \$44 would execute immediately because the stock price is less than \$44.

For Further Reference:

(Study Session 12, Modules 36.2, 36.3, LOS 36.e, 36.h)

CFA® Program Curriculum, Volume 4, page 240

CFA® Program Curriculum, Volume 4, page 246

Question #85 of 120

Question ID: 1214860

A multi-market index is *most likely* to overweight countries that have recently experienced high equity market returns if it weights the securities in each country index by:

- A) dividend yield, and weights the country index returns by market capitalization.**
- B) dividend yield, and weights the country index returns by gross domestic product.
- C) market capitalization, and weights the country index returns by gross domestic product.

Explanation

Weighting countries in a multi-market index by market capitalization, rather than a fundamental factor such as GDP, is likely to overweight countries with previously high stock returns.

For Further Reference:

(Study Session 12, Module 37.2, LOS 37.h)

CFA® Program Curriculum, Volume 4, page 297

Question #86 of 120

Question ID: 1214862

An analyst with Guffman Investments has developed a stock selection model based on earnings announcements made by companies with high P/E stocks. The model predicts that investing in companies with P/E ratios twice that of their industry average that make positive earnings announcements will generate significant excess return. If the analyst has consistently made superior risk-adjusted returns using this strategy, which form of the efficient market hypothesis has been violated?

- A) Weak form only.
- B) Semistrong and strong forms only.**
- C) Strong, semistrong, and weak forms.

Explanation

The semistrong form of EMH states that security prices rapidly adjust to reflect all publicly available information. If the analyst can use his model, which is based on publicly available information, to earn above average returns, the semistrong form of the EMH has been violated. If the semistrong form of EMH is violated, the strong form of EMH is also violated.

For Further Reference:

(Study Session 12, Module 38.1, LOS 38.d, 38.e)

CFA® Program Curriculum, Volume 4, page 326

CFA® Program Curriculum, Volume 4, page 330

Question #87 of 120

Question ID: 1214873

Questions 87 through 99 relate to Fixed Income. (19.5 minutes)

An analyst collects the following spot rates, stated as annual BEYs:

- 6-month spot rate = 6%.
- 12-month spot rate = 6.5%.
- 18-month spot rate = 7%.
- 24-month spot rate = 7.5%.

Given only this information, the price of a 2-year, semiannual-pay, 10% coupon bond with a face value of \$1,000 is *closest* to:

- A) \$918.30.
- B) \$1,000.00.
- C) \$1,046.77.

Explanation

This question can be answered without calculations. Since the spot rates are less than the coupon rate, the price must be greater than par value, so C is the only possible correct choice.

This is a four-period bond with \$50 cash flows each period. Divide each spot rate by two to get the semiannual rate.

$$PV_1: N = 1; I/Y = 3.00; FV = \$50; CPT \rightarrow PV = \$48.54$$

$$PV_2: N = 2; I/Y = 3.25; FV = \$50; CPT \rightarrow PV = \$46.90$$

$$PV_3: N = 3; I/Y = 3.50; FV = \$50; CPT \rightarrow PV = \$45.10$$

$$PV_4: N = 4; I/Y = 3.75; FV = \$1,050; CPT \rightarrow PV = \$906.23$$

Sum to get \$1,046.77.

$$\text{Or, } \frac{50}{1.03} + \frac{50}{1.0325^2} + \frac{50}{1.035^3} + \frac{1,050}{1.0375^4} = \$1,046.77$$

For Further Reference:

(Study Session 14, Module 44.2, LOS 44.c)

CFA® Program Curriculum: Volume 5, page 117

Question #88 of 120

Question ID: 1214880

A bond has an effective duration of 7.5. If the bond yield changes by 100 basis points, the price of the bond will change by:

- A) exactly 0.75%.
- B) approximately 7.5%.**
- C) approximately 0.75%.

Explanation

The change in price due to a change in yield is only approximate because the calculation of effective duration does not reflect all of the curvature of the price-yield curve (convexity). It is a linear approximation of a non-linear relation.

For Further Reference:

(Study Session 15, Module 46.1, LOS 46.b)

CFA® Program Curriculum: Volume 5, page 248

Question #89 of 120

Question ID: 1214878

Which of the following *best* describes the motivation for a corporation to issue securitized bonds? Securitization of specific assets by a corporation enables the corporation to:

- A) improve the recovery rate in the event of default.
- B) use the assets as collateral for additional borrowing.
- C) get a credit rating on the bonds that will result in a lower cost of borrowing.**

Explanation

Issuing securitized bonds from a special purpose entity allows a corporation to dedicate the assets' cash flows to specific debt issues. This enables the issue to receive a higher credit rating than that of the corporation.

For Further Reference:

(Study Session 14, Modules 42.1, 45.1, LOS 42.d, 45.b)

CFA® Program Curriculum: Volume 5, page 21

CFA® Program Curriculum: Volume 5, page 181

Question #90 of 120

Question ID: 1214871

Which of the following is a risk faced by issuers of commercial paper?

- A) Default risk.
- B) Rollover risk.**
- C) Reinvestment risk.

Explanation

Rollover risk is the risk that an issuer who relies on the commercial paper market as a funding source may not be able to issue new commercial paper when an outstanding issue matures. Default risk and reinvestment risk are faced by bondholders.

For Further Reference:

Question #91 of 120

Question ID: 1214881

Other things equal, an increase in an option-free bond's yield to maturity will:

- A) increase its interest rate risk.
- B) decrease its interest rate risk.**
- C) not change its interest rate risk.

Explanation

Because the price-yield relationship for an option-free bond is convex, interest rate risk as measured by duration changes when a bond's YTM changes. An increase in YTM reaches a flatter part of the price-yield curve, from which changes in yield will have relatively smaller effects on the bond's value.

For Further Reference:

(Study Session 15, Module 46.2, LOS 46.e)

CFA® Program Curriculum: Volume 5, page 259

Question #92 of 120

Question ID: 1214876

An analyst collects the following information regarding spot rates:

- 1-year rate = 4%.
- 2-year rate = 5%.
- 3-year rate = 6%.
- 4-year rate = 7%.

The 2-year forward rate two years from today is *closest* to:

- A) 7.02%.**
- B) 8.03%.
- C) 9.04%.

Explanation

$$\sqrt{\frac{(1.07)^4}{(1.05)^2}} - 1 = 0.0904, \text{ or } \frac{(4 \times 7) - (2 \times 5)}{2} = 9 \text{ as an approximation.}$$

For Further Reference:

(Study Session 14, Module 44.4, LOS 44.j)

CFA® Program Curriculum: Volume 5, page 143

Question #93 of 120

Question ID: 1214884

A credit analyst determines the following selected financial ratios for three firms in an industry after making all appropriate adjustments:

	Johnson	Knight	Lawrence
EBIT/revenue	0.25	0.30	0.35
EBIT/interest	12.5	14.0	10.5
Debt/capital	0.55	0.70	0.35
Revenue/assets	0.20	0.15	0.10

In evaluating the creditworthiness of these firms, the analyst should conclude that:

- A) Knight has the most favorable leverage and Johnson has the most favorable coverage.
- B) Lawrence has the most favorable leverage and Knight has the most favorable coverage.**
- C) Johnson has the most favorable leverage and Lawrence has the most favorable coverage.

Explanation

EBIT / interest is a coverage ratio and debt / capital is a leverage ratio. Higher interest coverage and lower leverage are favorable for creditworthiness. Of the three companies given, Knight has the highest interest coverage and Lawrence has the lowest leverage. EBIT / revenue (operating profit margin) is a profitability ratio and revenue / assets (asset turnover) is an operating ratio.

For Further Reference:

(Study Session 7, Module 24.3, LOS 24.c; Study Session 8, Module 28.6, LOS 28.k; Study Session 15, Module 47.2, LOS 47.g, 47.h)

CFA® Program Curriculum: Volume 5, page 325

CFA® Program Curriculum: Volume 5, page 325

Question #94 of 120

Question ID: 1214882

How should an analyst interpret a downward-sloping term structure of yield volatility? Short-term interest rates are:

- A) higher than long-term interest rates.
- B) more stable than long-term interest rates.
- C) more variable than long-term interest rates.**

Explanation

The term structure of yield volatility refers to the volatility of interest rates at different maturities. If the term structure of yield volatility is downward sloping, short-term interest rates are more volatile than long-term interest rates.

For Further Reference:

(Study Session 15, Module 46.3, LOS 46.j)

CFA® Program Curriculum: Volume 5, page 278

Question #95 of 120

Question ID: 1214872

Parsons Inc. is issuing an annual-pay bond that will pay no coupon for the first five years and then pay a 10% coupon for the remaining five years to maturity. The 10% coupon interest for the first five years will all be paid (without additional interest) at maturity. If the annual YTM on this bond is 10%, the price of the bond per \$1,000 of face value is *closest* to:

- A) \$778.
- B) \$814.
- C) \$856.

Explanation

This bond has no cash flows for the first five years. It then has a \$100 cash flow for years 6 through 10. Additionally, the accrued interest (\$500) that wasn't paid in the first five years would have to be paid at the end, along with the principal. A financial calculator using the CF/NPV worksheet can handle this type of problem. The required inputs are $CF_0 = 0$, $CF_1 = 0$, $F_1 = 5$, $CF_2 = 100$, $F_2 = 4$, $CF_3 = 1,600$, $F_3 = 1$, NPV, $I = 10\%$, CPT = 813.69. Note that CF_3 is made up of the principal (\$1,000) plus the remaining \$100 coupon plus the accrued interest (\$500) that was not paid during the first five years of the bond's life.

For Further Reference:

(Study Session 14, Module 44.1, LOS 44.a)

CFA® Program Curriculum: Volume 5, page 108

Question #96 of 120

Question ID: 1214875

For a 5-year pure discount bond that is callable at par after two years, which of the following yield measures will be lowest?

- A) Current yield.
- B) Yield to first call.
- C) Yield to maturity.

Explanation

Current yield = annual coupon / bond price. A pure discount bond is a zero-coupon bond, which has a current yield of zero.

For Further Reference:

(Study Session 14, Module 44.3, LOS 44.g)

CFA® Program Curriculum: Volume 5, page 126

Question #97 of 120

Question ID: 1214879

What is *most likely* to happen to the prepayment rate and the weighted average life of a typical pass-through security if mortgage rates decrease?

- A) Both will increase.
- B) Both will decrease.
- C) One will increase and one will decrease.

Explanation

Prepayment rates will most likely increase if mortgage rates decrease. Increasing prepayments will decrease the weighted average life of the pass-through security.

For Further Reference:

(Study Session 14, Module 45.2, LOS 45.e, 45.f)

CFA® Program Curriculum: Volume 5, page 194

CFA® Program Curriculum: Volume 5, page 194

Question #98 of 120

Question ID: 1214874

Accrued interest is the difference between a coupon bond's:

- A) flat price and full price.**
- B) clean price and flat price.
- C) invoice price and dirty price.

Explanation

Flat price or clean price does not include accrued interest. Full price (dirty or invoice price) includes accrued interest.

For Further Reference:

(Study Session 14, Module 44.2, LOS 44.d)

CFA® Program Curriculum: Volume 5, page 119

Question #99 of 120

Question ID: 1214883

In what way is approximate convexity different from effective convexity?

- A) Effective convexity takes embedded options into account, while approximate convexity does not.**
- B) Effective convexity results in a more accurate estimate of an option-free bond's change in price than approximate convexity.
- C) Approximate convexity can be used with an unequal increase and decrease in yield, while effective convexity can only be used with an equal increase or decrease in yield.

Explanation

The distinction between effective convexity and approximate convexity is that effective convexity accounts for the effects of embedded options on the bond's cash flows at different yields. For bonds that do not have embedded options, there is no difference between effective convexity and approximate convexity. Both measures are used to improve the estimate of the change in a bond's price for a given change in yield.

For Further Reference:

(Study Session 15, Module 46.3, LOS 46.h)

CFA® Program Curriculum: Volume 5, page 269

Question #100 of 120

Question ID: 1214887

Questions 100 through 105 relate to Derivatives. (10.5 minutes)

A programming error resulted in the issuance of a forward contract that ignored the convenience yield investors realize from holding the underlying asset. To gain an arbitrage profit from this error, an arbitrageur's position would *most likely* include a long position in the:

- A) forward contract.
- B) underlying asset.**
- C) risk-free asset.

Explanation

Ignoring the convenience yield (a nonmonetary benefit of holding the asset) will result in a forward price that is above the no-arbitrage forward price. To profit from this mispricing, an arbitrageur would borrow at the risk-free rate (i.e., take a short position in the risk-free asset), purchase the underlying asset, and take a short position in the forward contract.

For Further Reference:

(Study Session 16, Module 49.1, LOS 49.a, 49.d)

CFA® Program Curriculum, Volume 5, page 456

CFA® Program Curriculum, Volume 5, page 469

Question #101 of 120

Question ID: 1214889

Compared to an otherwise identical European put option, one that has a longer time to expiration:

- A) may be worth less than the put that is nearer to expiration.**
- B) must be worth more than the put that is nearer to expiration.
- C) must be worth at least as much as the put that is nearer to expiration.

Explanation

Normally, options with greater time to expiration are worth more than otherwise identical options that are nearer to expiration. However, in some circumstances, this relationship may not hold for European puts. For example, if the price of the underlying asset goes to zero, the European put with less time to expiration may be worth more because the put holder will receive the exercise price earlier.

For Further Reference:

(Study Session 16, Module 49.3, LOS 49.k)

CFA® Program Curriculum, Volume 5, page 482

Question #102 of 120

Question ID: 1214891

An analyst using a one-period binomial model calculates a probability-weighted average of an option's values following an up-move and a down-move. According to this model, this average is *most likely*:

- A) equal to the option's value today.
- B) less than the option's value today.**

C) greater than the option's value today.

Explanation

The probability-weighted average calculated by the analyst is the option's value after one period. To estimate the option's value today, this result must be discounted by one period.

For Further Reference:

(Study Session 16, Module 49.4, LOS 49.n)

CFA® Program Curriculum, Volume 5, page 496

Question #103 of 120

Question ID: 1214888

Market conditions have increased the storage costs and decreased the convenience yield of a commodity. Other things equal, what effects will these changes have on the value of a forward contract on that commodity? The forward contract value:

- A) will increase.**
- B) will decrease.**
- C) may increase or decrease.**

Explanation

The value of a forward contract at any time before its expiration is the spot price of the underlying asset, plus the present value of storage costs, minus the present value of monetary and nonmonetary benefits of holding the underlying, minus the present value of the forward price. An increase in storage costs and a decrease in convenience yield (a non-monetary benefit of holding the asset) will increase the value of a forward contract, other things equal.

For Further Reference:

(Study Session 16, Module 49.1, LOS 49.e)

CFA® Program Curriculum, Volume 5, page 471

Question #104 of 120

Question ID: 1214885

Which of the following statements *most accurately* represents the positions of the parties to a derivatives contract?

- A) A call option imposes an obligation to buy the underlying security.**
- B) A forward contract imposes an obligation on the seller but not the buyer.**
- C) Both a put writer and a call writer have an obligation to exchange the underlying asset.**

Explanation

Both put and call writers (sellers) have an obligation to honor the terms of the option if it is exercised. Option holders (buyers) have the right, not the obligation, to exercise under the terms of the agreement. The holder pays a premium for this right, while the writer receives a premium for this obligation. A forward contract imposes an obligation on both the buyer and seller to exchange the underlying asset.

For Further Reference:

(Study Session 16, Module 48.1, LOS 48.b)

Question #105 of 120

Question ID: 1214890

Given the put-call parity relationship, a synthetic underlying asset can be created by forming a portfolio of a:

- A) long call, long put, and short risk-free bond.
- B) long call, short put, and long risk-free bond.**
- C) short call, long put, and long risk-free bond.

Explanation

An asset underlying put and call options can be replicated with a long European call option, a short European put option, and a long position in a risk-free bond that pays the exercise price on the expiration date.

For Further Reference:

(Study Session 16, Module 49.3, LOS 49.I)

CFA® Program Curriculum, Volume 5, page 478

Question #106 of 120

Question ID: 1214886

In futures markets, the primary role of the clearinghouse is to:

- A) prevent arbitrage and enforce federal regulations.
- B) act as guarantor to both sides of a futures trade.**
- C) reduce transaction costs by making contract prices public.

Explanation

Acting as the counterparty for all buyers and sellers is the primary role of the clearinghouse. By providing liquidity, the clearinghouse may also help lower transaction costs indirectly.

For Further Reference:

(Study Session 16, Module 48.1, LOS 48.c)

CFA® Program Curriculum, Volume 5, page 394

Question #107 of 120

Question ID: 1214892

Questions 107 through 113 relate to Alternative Investments. (10.5 minutes)

Which of the following characteristics *most likely* applies to private equity investing?

- A) Liquid secondary market.
- B) High degree of reliance on manager skill.**
- C) Relatively short-term investment horizon.

Explanation

Returns on private equity investments depend to a large extent on the skill of the general partners. Private equity investments are illiquid and require a long-term investment horizon.

For Further Reference:

(Study Session 17, Module 50.1, LOS 50.b)

CFA® Program Curriculum, Volume 6, page 10

Question #108 of 120

Question ID: 1214893

Compared to traditional investment managers who invest in long-only stocks and bonds, alternative investment managers typically invest in assets that are:

- A) less regulated, less transparent, and less liquid.**
- B) more transparent, less liquid, and less correlated with traditional investments.
- C) less transparent, more regulated, and less correlated with traditional investments.

Explanation

Compared to traditional investments, alternative investments exhibit less liquidity, less regulation, and less transparency. As a group, alternative investments generally have relatively low return correlations with traditional investments.

For Further Reference:

(Study Session 17, Module 50.1, LOS 50.a)

CFA® Program Curriculum, Volume 6, page 6

Question #109 of 120

Question ID: 1214894

Supplying capital to companies that are just moving into operation, but do not as yet have a product or service available to sell, is a description that *best* relates to which of the following stages of venture capital investing?

- A) Early stage.**
- B) Mezzanine stage.
- C) Angel investing stage.

Explanation

The description relates best to the early stage wherein the capital that is supplied helps speed up product development and also helps pay for the beginning of a marketing campaign.

For Further Reference:

(Study Session 17, Module 50.1, LOS 50.b)

CFA® Program Curriculum, Volume 6, page 10

Question #110 of 120

Question ID: 1214895

An equity hedge fund that uses technical analysis techniques to identify undervalued shares to buy and overvalued shares to sell short is *best* described as pursuing:

- A) a market neutral strategy.
- B) a special situations strategy.
- C) a quantitative directional strategy.**

Explanation

Quantitative directional strategies employ technical analysis and may have net long or short exposure. A market neutral strategy maintains approximately equal values in long and short positions and is typically based on fundamental analysis. Special situations strategy is an event-driven strategy that involves investing in firms that are selling divisions or assets, distributing capital, or issuing or repurchasing securities.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.b)

CFA® Program Curriculum, Volume 6, page 10

Question #111 of 120

Question ID: 1214896

A hedge fund that requires incentive fees to be calculated only on the portion of returns above a benchmark return is said to have a:

- A) soft hurdle rate.
- B) hard hurdle rate.**
- C) high water mark.

Explanation

In a hedge fund's fee structure, a hard hurdle rate means that incentive fees are earned only on returns in excess of the benchmark return. A soft hurdle rate means that incentive fees are calculated on the entire return, but are only paid if the return exceeds the hurdle rate. A high water mark specifies that incentive fees are only paid on returns that increase an investor's account value above its highest previous value.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.d)

CFA® Program Curriculum, Volume 6, page 20

Question #112 of 120

Question ID: 1214897

Compared to investing in commodities, investing in farmland is *most likely* to provide which of the following benefits?

- A) More liquidity.
- B) Higher income.**
- C) Better inflation hedge.

Explanation

Investing in farmland may provide income over the life of the investment from crops produced and sold. Investing in commodities typically does not provide income. Commodities and real estate investments such as farmland may both provide a hedge against inflation. Direct investments in real estate are less liquid than investments in commodities, which are typically made through highly liquid exchange-traded derivatives.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.b)

CFA® Program Curriculum, Volume 6, page 10

Question #113 of 120

Question ID: 1214898

An investment in commodities will *most likely* provide returns from:

- A) cash flows from the assets.
- B) the yield on collateral deposits.
- C) changes in commodity spot prices.**

Explanation

Returns from holding commodities depend on changes in spot prices. Investing directly in commodities requires no collateral deposit; this is required for investing in commodity derivatives. Commodities typically do not generate cash flows.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.b)

CFA® Program Curriculum, Volume 6, page 10

Question #114 of 120

Question ID: 1214903

Questions 114 through 120 relate to Portfolio Management. (10.5 minutes)

In the context of the capital asset pricing model, an active manager would be *most likely* to purchase a security that plots:

- A) on the security market line.
- B) below the security market line.
- C) above the security market line.**

Explanation

Active management seeks to identify mispriced assets, which are likely to earn abnormal returns. A security that plots above the security market line (SML) is undervalued (i.e., expected to earn a return that is higher than it would if it were priced at its equilibrium price, given its systematic risk).

For Further Reference:

(Study Session 18, Module 53.2, LOS 53.h)

CFA® Program Curriculum, Volume 6, page 217

Question #115 of 120

Question ID: 1214899

The investment needs of property and casualty insurers are characterized by a:

- A) short-term time horizon and low risk tolerance.**
- B) long-term time horizon and high risk tolerance.
- C) short-term time horizon and high risk tolerance.

Explanation

Property and casualty insurers typically have a short-term time horizon, since claims are expected in the near term, and low risk tolerance.

For Further Reference:

(Study Session 18, Module 51.1, LOS 51.b)

CFA® Program Curriculum, Volume 6, page 89

Question #116 of 120

Question ID: 1214900

Using historical index returns for an equities market over a 20-year period, an analyst has calculated the average annual return as 5.60% and the holding period return as 170%. The compound annual index return over the period is *closest* to:

- A) 2.69%.**
- B) 5.09%.
- C) 5.24%.

Explanation

$$(1 + \text{HPR})^{1/n} - 1 = (1 + 1.70)^{1/20} - 1 = 0.050916$$

For Further Reference:

(Study Session 18, Module 52.1, LOS 52.a)

CFA® Program Curriculum, Volume 6, page 116

Question #117 of 120

Question ID: 1214904

Which portion of an investment policy statement is *most likely* to state any restriction on portfolio leverage?

- A) Procedures.**
- B) Investment guidelines.
- C) Duties and responsibilities.

Explanation

The investment guidelines section typically contains information about specific types of assets prohibited in the portfolio and the permissible use, if any, of leverage and derivatives.

For Further Reference:

Question #118 of 120

Question ID: 1214901

When a risk-free asset is combined with a portfolio of risky assets, which of the following is *least accurate*?

- A) The standard deviation of the return for the newly created portfolio is the standard deviation of the returns of the risky asset portfolio multiplied by its portfolio weight.
- B) The expected return for the newly created portfolio is the weighted average of the return on the risk-free asset and the expected return on the risky asset portfolio.
- C) **The variance of the resulting portfolio is a weighted average of the returns variances of the risk-free asset and of the portfolio of risky assets.**

Explanation

This statement is not correct; the standard deviation of returns for the resulting portfolio is a weighted average of the returns standard deviation of the risk-free asset (zero) and the returns standard deviation of the risky-asset portfolio.

For Further Reference:

(Study Session 18, Module 53.1, LOS 53.a)

CFA® Program Curriculum, Volume 6, page 192

Question #119 of 120

Question ID: 1214902

If a stock's beta is equal to 1.2, its standard deviation of returns is 28%, and the standard deviation of the returns on the market portfolio is 14%, the covariance of the stock's returns with the returns on the market portfolio is *closest* to:

- A) 0.168.
- B) **0.024.**
- C) 0.600.

Explanation

From the fact that $\beta_i = \text{Cov}_{i,\text{mkt}} / \text{Var}_{\text{mkt}}$, we have $\text{Cov}_{i,\text{mkt}} = \beta_i \times \text{var}_{\text{mkt}}$.

$$\text{Cov}_{i,\text{mkt}} = 1.2 \times 0.14^2 = 0.02352.$$

For Further Reference:

(Study Session 18, Module 53.1, LOS 53.e)

CFA® Program Curriculum, Volume 6, page 206

Question #120 of 120

Question ID: 1214905

Non-financial sources of risk for an organization *most likely* include:

- A) credit risk.
- B) liquidity risk.
- C) **solvency risk.**

Explanation

Non-financial risks are those that arise from an organization's operations and from outside sources other than financial markets. Solvency risk is classified as a non-financial risk.

For Further Reference:

(Study Session 19, Module 55.1, LOS 55.f)

CFA® Program Curriculum, Volume 6, page 307